



Revisiting Graph-Level Anomaly Detection: From Partially to Fully Unsupervised Learning

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Abstract

Graph-level anomaly detection (GLAD) is a critical task to identify graphs with abnormal properties in various domains, ranging from fraudulent social networks to malicious botnets on online platforms. The dominant paradigm for existing GLAD detectors has been partially unsupervised, relying on training data composed exclusively of normal samples. However, this partially unsupervised paradigm inevitably requires a costly expert filtering process to ensure the training data is free of anomalies. This creates a significant gap between current approaches and the real-world necessity of a fully unsupervised paradigm, which involves training a model directly on real-world data "as-is", with its inherent mix of normal and anomalous samples. To bridge this gap, we incorporate uncertainty learning into GLAD to promote fully unsupervised learning. We propose two frameworks: Score Uncertainty Learning (SUL) and Graph-data Uncertainty Learning (GUL) ¹. Specifically, SUL enhances existing GLAD detectors by modeling uncertainty through Gaussian distributions over the detectors' predictions, adaptively attenuating the influence of potential anomalies. GUL is an end-to-end framework that iteratively optimizes anomaly detection and uncertainty modeling via an Expectation-Maximization algorithm. In addition, we develop a dedicated loss that utilizes potential anomalies to enhance the effectiveness and robustness of GUL. Empirical results on sixteen benchmark datasets, covering real-world graphs

from social networks and online platforms, demonstrate the superiority of our methods and highlight the promise of incorporating uncertainty into fully unsupervised GLAD.

CCS Concepts

• Information systems → Data mining.

Keywords

Graph-level Anomaly Detection, Uncertainty Estimation

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The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18333850>.

1 Introduction

Within a set of graphs, rare individuals could exhibit abnormal properties/patterns that deviate from others — referred to as graph-level anomalies — such as fraudulent communities on e-commerce platforms [27, 55] and networks spreading fake news [11, 33]. Identifying these anomalies offers significant value in scanning potential threats and facilitating real-world applications, e.g., financial fraud detection and disinformation tracking. Hence, Graph-level Anomaly Detection (GLAD) has become a critical task.

Recently, GLAD has attracted growing interest from the deep graph learning community. Several researchers tackle this task in a supervised manner [10, 28, 32, 43, 59] with achieving promising results. However, their practical application is limited by the scarcity of anomalous labels; for instance, disordered brain networks are not only naturally rare in medical databases, but also

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¹Our code is available at <https://github.com/ZhenyuYangMQ/UncertaintyGLAD>.



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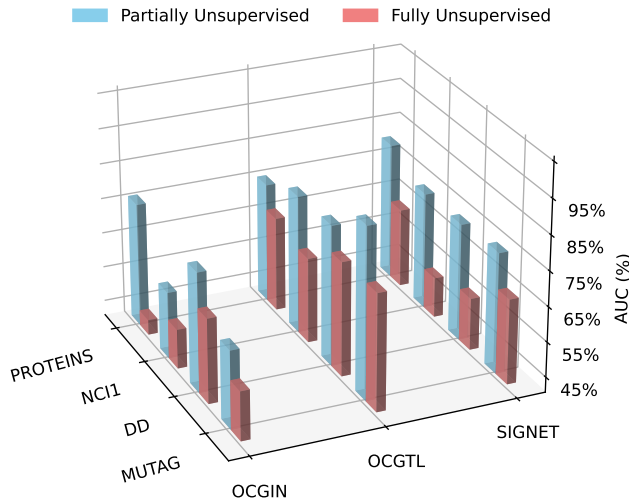


Figure 1: Performance of GLAD detectors (OCCIN, OCCGL, SIGNET) on four benchmark datasets under partially and fully unsupervised settings.

require significant domain expertise to annotate. To approach realistic deployment, an alternative paradigm has become prevalent: **training a deep model with solely "normal" samples**. This paradigm aims to leverage normal samples to tell the model what "normal" data look like (i.e., normality), thereby identifying anomalies through deviations from learned normality [42]. Consequently, this has established a dominant paradigm best described as partially unsupervised GLAD², which has led to the rapid development of various detectors [4, 5, 22, 30, 31, 40, 53, 60].

However, a discernible gap still exists between this partially unsupervised paradigm and fully unsupervised real-world scenarios: its success heavily relies on training with entirely "normal" data to establish a reliable normality. Nevertheless, this reliance on entirely "normal" data imposes a considerable overhead, as it demands a costly expert filtering of the training set; even more critically, it falters when faced with novel or unknown anomalies. On the other hand, training models from this partially unsupervised paradigm directly on the data "as-is" - with its inherent mix of normal and anomalous samples - inevitably corrupts the learned normality, leading to significant performance degradation (illustrated in Fig. 1). In applications such as detecting fake news or financial fraud, assuming that the collected data is entirely "normal" can result in substantial financial losses due to detection failures. For example, the \$10 billion losses as the Federal Trade Commission reported in 2023³. Thus, advancing GLAD from a partially to a fully unsupervised paradigm is critical for real-world applications.

²We call this "partially unsupervised" because, while training without explicit label information, it relies on implicit supervision—expert filtering for an entirely "normal" set — unlike a fully unsupervised approach where data is used "as-is".

³<https://www.ftc.gov/news-events/news/press-releases/2024/02/nationwide-fraud-losses-top-10-billion-2023-ftc-steps-efforts-protect-public>

This gap in GLAD tasks has been less explored, with the limited potential solutions being a few general efforts [41, 54] originally developed for image and tabular data. Specifically, Refine [54] iteratively filters out potential anomalies in the training set based on detectors' predictions, while LOE [41] further introduced a specialized loss to utilize these filtered potential anomalies. However, such strategies are unsatisfactory for GLAD tasks. A key limitation is that determining the amount of data to filter out requires prior knowledge of the anomaly count within the training set. More critically, their filtering relies on off-the-shelf detectors – which tend to produce unreliable results when trained on truly unsupervised scenarios (see Fig. 1)– risks further exacerbating biases.

Uncertainty, as a measure of a model's confidence in its predictions or decisions for a given sample [13, 20], offers a more promising direction than previous efforts. Intuitively, as GLAD detectors aim to learn the normality shared by normal samples, anomalies tend to bring uncertain statistical shifts. This quantified uncertainty is flexible for handling potential anomalies, eliminating the need for deterministic filtering. Despite this promise, this endeavor is fraught with two coupled challenges: *C1. How to model uncertainty in GLAD?* Existing uncertainty learning [7, 20] derives uncertainty via deviations between predictions and labels. In contrast, the GLAD task lacks ground-truth labels, relying instead on learned normality to infer whether samples are normal or anomalous. Thus, modeling uncertainty in the context of normality remains a complex and unresolved issue. *C2. How to incorporate uncertainty into GLAD?* Normality serves as the sole reference for both anomaly detection and uncertainty estimation. However, unlabeled anomalies degrade the quality of learned normality, necessitating uncertainty to guide and refine normality learning.

In this paper, we propose two varieties of uncertainty learning, i.e., **Score Uncertainty Learning (SUL)** and **Graph-data Uncertainty Learning (GUL)**, to enable truly unsupervised GLAD. SUL refines the existing GLAD detectors by estimating the Gaussian distribution over the detectors' predictions to model the uncertainty for each sample. The learned uncertainty can adaptively attenuate the impacts of potential anomalies. To tackle the concern that existing GLAD detectors fail to provide accurate predictions, we propose the GUL framework to direct model uncertainty from graph data. We formulate GUL as an Expectation–Maximization (EM) algorithm [34] that iteratively seeks the most representative graphs to characterize normality and optimize network parameters to model uncertainty. Specifically, the E-step estimates several representative graphs with the lowest uncertainty values to describe normality, supported by theoretical proof (*For C2*). In the M-step, the network parameters are optimized by maximizing the likelihood between training samples and estimated normality-representative graphs (*For C1*). In addition, an auxiliary loss is proposed to separate normality-representative graphs and potential anomalies. The main contributions of this work are:

- Our SUL framework demonstrates uncertainty's promise, outperforming prior efforts on fully unsupervised GLAD with a simple attenuation of potential anomalies.
- We develop GUL, an end-to-end framework that iteratively optimizes anomaly detection and uncertainty modeling through an EM algorithm.

- Extensive experiments on sixteen benchmark datasets demonstrate the superiority of incorporating uncertainty to pursue fully unsupervised GLAD.

2 Preliminaries and Related Work

This section outlines the preliminaries and briefly reviews related work. A more detailed literature review is available in Appendix A. **Notations.** Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote a graph with node set $\mathcal{V}$ and edge set $\mathcal{E}$, where each node $v \in \mathcal{V}$ has an attribute vector $\mathbf{x}_v$. The topology of $\mathcal{G}$ is represented by an adjacency matrix $\mathbf{A} \in \{0, 1\}^{n \times n}$, in which $A_{u,v} = 1$ indicates that there is an edge between nodes u and v , i.e., $\mathcal{E}_{u,v} \in \mathcal{E}$. We use $\mathbb{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_N\}$ to denote the graph set consisting of N graphs.

Graph Neural Networks (GNNs). GNNs [23, 46, 49, 51] can be utilized as the encoder to map the graph $\mathcal{G}$ into a vectorized representation $\mathbf{h}_{\mathcal{G}}$. First, node embeddings $\mathbf{h}_v$ are generated by GNNs based on node attribute $\mathbf{x}_v$ and structure information in $\mathbf{A}$. Then, the GNNs summarize all node embeddings in $\mathcal{G}$ to represent the graph as a vectorized representation $\mathbf{h}_{\mathcal{G}}$ through the readout function.

Data Uncertainty Learning. This work focuses on data uncertainty, which stems from inherent noise or ambiguity in data observations [1, 13]. In the context of fully unsupervised GLAD, data uncertainty primarily arises from unlabeled anomalies, which introduce uncertain statistical shifts during normality learning. Several prior works [25, 36] propose capturing data uncertainty as an input-dependent variable. Specifically, the data uncertainty is captured by modeling the variance $\sigma^2(x_i)$ as the output of a neural network, with each input x_i . Assuming that the label y_i follows a Gaussian distribution conditioned on the model prediction $\hat{y}_i$ and the variance $\sigma^2(x_i)$, i.e., $p(y_i | \hat{y}_i, \sigma^2(x_i)) = \mathcal{N}(y_i; \hat{y}_i, \sigma^2(x_i))$, the optimization of $\sigma^2(x_i)$ is trained by minimizing the negative log-likelihood $-\log p(y_i | \hat{y}_i, \sigma^2(x_i))$.

Partially Unsupervised GLAD. Partially Unsupervised GLAD detectors [22, 31, 35, 40, 58, 60] train an anomaly scoring function $\mathcal{F}_{\phi} : \mathcal{G} \rightarrow \mathbb{R}$ with parameter ϕ on a set of normal graphs to measure the degree of abnormality for any unseen graph. However, their success requires a training set consisting entirely of normal graphs.

Problem Formulation. In this paper, we target the fully unsupervised GLAD setting, aiming to train a detector $\mathcal{F}_{\phi}(\mathcal{G})$ directly on data "as-is", a mixed training set of both normal and anomalous graphs, where $\mathcal{F}_{\phi}(\mathcal{G}) : \mathcal{G} \rightarrow \mathbb{R}$ outputs the anomaly score. We further learn uncertainty $\sigma_{\theta}^2(\mathcal{G})$ with parameter θ to enhance the performance of GLAD detectors under fully unsupervised settings, where $\sigma_{\theta}^2(\mathcal{G}) : \mathcal{G} \rightarrow \mathbb{R}$ refers to the uncertainty of the model decision for $\mathcal{G}$. In the GUL framework, we simplified the network by retaining solely $\sigma_{\theta}^2(\mathcal{G})$, as the learned uncertainty is sufficiently accurate to be directly used as anomaly scores.

3 Enhancing Off-the-shelf Detectors with Score Uncertainty Learning (SUL)

In this section, we introduce Score Uncertainty Learning (SUL), a straightforward yet effective framework designed to enhance existing off-the-shelf GLAD detectors. SUL models uncertainty based on the anomaly scores provided by these detectors. The resulting uncertainties then act as adaptive scaling weights, which attenuate the influence of potential anomalies during the training

process. Unlike previous strategies [41, 54] that require presetting the number of potential anomalies for filtering, SUL adaptively attenuates the loss of potential anomalies through uncertainty.

In most GLAD detectors [26, 31, 35, 40, 60], the anomaly score $\mathcal{F}_{\phi}(\mathcal{G})$ closer to 0 indicates the higher likelihood that $\mathcal{G}$ is considered a normal graph. The training objective of a standard GLAD detector is to minimize $\mathcal{F}_{\phi}(\mathcal{G})$ to 0 in the partially unsupervised setting, since the training set consists of entirely normal samples. In the fully unsupervised scenario, this training objective can be misleading due to the inclusion of unlabeled anomalies. Thus, we incorporate uncertainty to attenuate this objective on constraining potential anomalies. To model the score uncertainty, we assume a Gaussian distribution as $p(0 | \mathcal{G}, \phi, \theta) \sim \mathcal{N}(0; \mathcal{F}_{\phi}(\mathcal{G}), \sigma_{\theta}^2(\mathcal{G}))$. Given that normal samples make up the majority, we maximize the following likelihood,

$$p(0 | \mathcal{G}, \phi, \theta) = \frac{1}{\sqrt{2\pi\sigma_{\theta}^2(\mathcal{G})}} \exp\left(-\frac{(0 - \mathcal{F}_{\phi}(\mathcal{G}))^2}{2\sigma_{\theta}^2(\mathcal{G})}\right), \quad (1)$$

Subsequently, we minimize the negative log-likelihood,

$$\mathcal{L}_{SUL} = -\log p(0 | \mathcal{G}, \phi, \theta) = \frac{\log \sigma_{\theta}^2(\mathcal{G})}{2} + \frac{\mathcal{F}_{\phi}^2(\mathcal{G})}{2\sigma_{\theta}^2(\mathcal{G})}, \quad (2)$$

where $\mathcal{F}_{\phi}(\mathcal{G})$ could be an off-the-shelf anomaly scoring function, and $\sigma_{\theta}^2(\mathcal{G})$ is modeled by neural networks. In practice, we train a GNN plus a Multilayer Perceptron (MLP) to predict log variance: $\log \sigma_{\theta}^2(\mathcal{G})$. Consequently, the final anomaly score is obtained as $\exp(\log \sigma_{\theta}^2(\mathcal{G}))$, which inherently avoids negative values.

Uncertainty Attenuation Mechanism. Since Eq. 2 targets an anomaly score of 0 for an arbitrary sample $\mathcal{G}$, this training objective naturally struggles to fit anomalies, inducing high uncertainty. Specifically, for those anomalies with higher anomaly scores²: $\mathcal{F}_{\phi}^2(\mathcal{G})$, SUL will estimate higher variances (i.e., uncertainty) to attenuate losses, referred to as $\frac{\mathcal{F}_{\phi}^2(\mathcal{G})}{2\sigma_{\theta}^2(\mathcal{G})}$. In addition, SUL is discouraged from predicting high variance for all samples, as the first term in Eq. 2: $\log \sigma_{\theta}^2(\mathcal{G})$, will regularize the model. Hence, SUL focuses more on samples with low anomaly scores while paying less attention to those with high ones that may be potential anomalies. Instead of filtering a fixed number of samples, SUL allows the model to adaptively attenuate impacts of potential anomalies during training.

Discussion: As a simplified but effective implementation, SUL can be plugged into a wide range of existing GLAD detectors without requiring a predefined number of potential anomalies for filtering. However, **the critical concern remains for SUL: If the anomaly scores produced by the GLAD detectors under the fully unsupervised setting are inaccurate, could the modeled uncertainty still be deemed reliable?** To pursue more accurate uncertainty modeling and improve the reliability of fully unsupervised anomaly detection, we propose directly modeling uncertainty from the graph data itself.

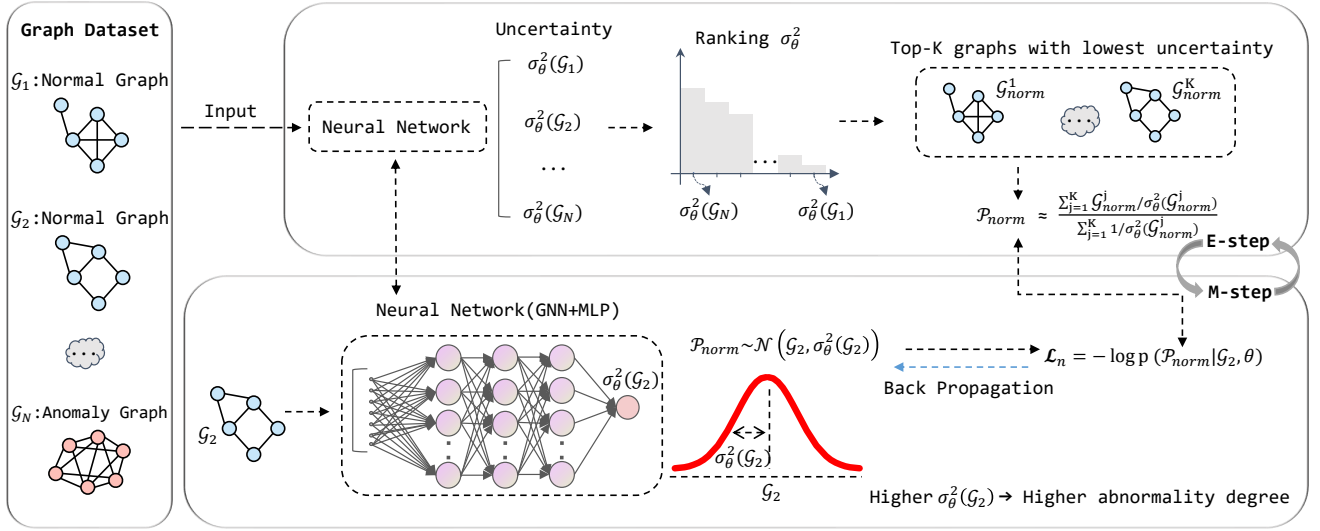


Figure 2: The overview of GUL framework involving E-step and M-step.

4 Our End-to-end Framework: Graph-data Uncertainty Learning (GUL)

4.1 Overview of Framework

By discussing the limitations of SUL, i.e., its reliance on potentially unreliable anomaly scores, we introduce our core contribution: Graph-data Uncertainty Learning (GUL), an end-to-end framework that learns uncertainty directly from graph data. At the heart of GUL lies an Expectation-Maximization (EM) algorithm, which, as illustrated in Fig. 2, iteratively alternates between two steps: an E-step to estimate a set of representative graphs that characterize normality, and an M-step to optimize the network for uncertainty modeling. By alternating between these two steps, GUL progressively refines both the normality estimation for distinguishing anomalies and the uncertainty model.

4.2 E-step: Estimating Normality-representative Graphs

Normality refers to stable patterns or properties shared by many normal graphs but rarely or never occur among anomalies. For convenience, we express normality as $\mathcal{P}_{norm}$ in this paper. Similar to Section 3, we assume the Gaussian distribution $\mathcal{P}_{norm} \sim \mathcal{N}(\mathcal{G}, \sigma_\theta^2(\mathcal{G}))$ around normality and data, to formulate,

$$p(\mathcal{P}_{norm} | \mathcal{G}, \theta) = \frac{1}{\sqrt{2\pi\sigma_\theta^2(\mathcal{G})}} \exp\left(-\frac{(\mathcal{P}_{norm} - \mathcal{G})^2}{2\sigma_\theta^2(\mathcal{G})}\right). \quad (3)$$

The network parameters θ are fixed in the E-step; thus, we can estimate the best $\mathcal{P}_{norm}$ by maximizing the posterior probability,

$$\arg \max_{\mathcal{P}_{norm}} p(\mathcal{P}_{norm} | \mathcal{G}, \theta) \implies \mathcal{P}_{norm} = \frac{\sum_{i=1}^N \mathcal{G}_i / \sigma_\theta^2(\mathcal{G}_i)}{\sum_{i=1}^N 1 / \sigma_\theta^2(\mathcal{G}_i)}. \quad (4)$$

At this stage, we have explicitly modeled the latent variable $\mathcal{P}_{norm}$ via all graphs and their uncertainties. This aligns with the

fundamental concept that normality presents shared patterns (or properties) among most graphs. However, using all graphs in the training set to compute $\mathcal{P}_{norm}$ will afford a high computational cost and introduce noise, as there are unavoidable anomalies within a fully unsupervised scenario. Therefore, we propose a sampling approximation strategy with further theoretical analysis.

Proposition. To approximate $\mathcal{P}_{norm}$, one can select K normality-representative graphs with the lowest variances $\sigma_\theta^2(\mathcal{G})$, denoted as $\{\mathcal{G}_{norm}^j\}_{j=1}^K$. This approximation is quantitatively formulated as a Truncated Weighted Mean, which minimizes the expected error in representing normality by exclusively utilizing low-uncertainty samples while discarding high-uncertainty noise in Eq. 4.

Proof. As shown in Eq. 4, the estimation of normality $\mathcal{P}_{norm}$ utilizing all graphs in the training set. We decompose this estimation into two disjoint parts: the low-uncertainty normality-representative set $\{\mathcal{G}_{norm}^j\}_{j=1}^K$ and the remaining set of high-uncertainty samples, as

$$\mathcal{P}_{norm} = \frac{\sum_{j=1}^K \mathcal{G}_{norm}^j / \sigma_\theta^2(\mathcal{G}_{norm}^j) + \sum_{\mathcal{G}_k \notin \{\mathcal{G}_{norm}^j\}_{j=1}^K} \mathcal{G}_k / \sigma_\theta^2(\mathcal{G}_k)}{\sum_{j=1}^K 1 / \sigma_\theta^2(\mathcal{G}_{norm}^j) + \sum_{\mathcal{G}_k \notin \{\mathcal{G}_{norm}^j\}_{j=1}^K} 1 / \sigma_\theta^2(\mathcal{G}_k)}. \quad (5)$$

This formula constitutes a weighted mean, and the inverse of variance $1/\sigma_\theta^2(\mathcal{G}_k)$ serves as the weight assigned to the graph $\mathcal{G}_k$. However, the training set inherently contains anomalies in the fully unsupervised setting. Incorporating these outliers in the weighted mean introduces bias in estimating normality $\mathcal{P}_{norm}$. Therefore, adopting a Truncated Weighted Mean is statistically more robust. Since the set $\{\mathcal{G}_{norm}^j\}_{j=1}^K$ contains graphs with the lowest variances, any graph $\mathcal{G}_k$ in the remaining set has a strictly larger variance $\sigma_\theta^2(\mathcal{G}_k)$ than those in $\{\mathcal{G}_{norm}^j\}_{j=1}^K$, resulting in a smaller weight. This renders the contribution of the remaining set negligible relative to the representative set, allowing for its truncation.

To sum up, the approximation of $\mathcal{P}_{norm}$ in the E-step is,

$$\mathcal{P}_{norm} \approx \frac{\sum_{j=1}^K \mathcal{G}_{norm}^j / \sigma_\theta^2(\mathcal{G}_{norm}^j)}{\sum_{j=1}^K 1 / \sigma_\theta^2(\mathcal{G}_{norm}^j)}, \quad (6)$$

where K is the hyper-parameter. In practice, we found that the GUL framework is robust to variations in K across a wide range of values, in Section 5.6.

4.3 M-step: Normality-based Uncertainty Learning

During the E-step and Eq. 6, the normality $\mathcal{P}_{norm}$ has been well studied by incorporating uncertainty. This section presents the M-step for learning uncertainty $\sigma_\theta^2(\mathcal{G})$ based on estimated normality $\mathcal{P}_{norm}$, where the goal is to optimize network parameters θ . By alternating between the E-step and M-step, the GUL iterative update normality $\mathcal{P}_{norm}$ and network parameters θ , respectively.

The uncertainty in GUL is learned by the neural network $\sigma_\theta^2(\mathcal{G})$ with parameters θ , which is fixed in the E-step. We revisit the assumption that $\mathcal{P}_{norm} \sim \mathcal{N}(\mathcal{G}, \sigma_\theta^2(\mathcal{G}))$, with the corresponding probability density $p(\mathcal{P}_{norm} | \mathcal{G}, \theta) = \frac{1}{\sqrt{2\pi\sigma_\theta^2(\mathcal{G})}} \exp\left(-\frac{(\mathcal{P}_{norm}-\mathcal{G})^2}{2\sigma_\theta^2(\mathcal{G})}\right)$, but the M-step holds $\mathcal{P}_{norm}$ constant and optimizes parameters θ by maximizing the likelihood. This process is refined to minimize the negative log-likelihood based on fixed $\mathcal{P}_{norm}$, denoted as,

$$\mathcal{L}_n = \frac{1}{2} \sum_{i=1}^N \left[\log \sigma_\theta^2(\mathcal{G}_i) + \frac{(\mathcal{P}_{norm} - \mathcal{G}_i)^2}{\sigma_\theta^2(\mathcal{G}_i)} \right]. \quad (7)$$

Plugging the approximation from Eq. 6 into Eq. 7, the loss function is reformulated as,

$$\mathcal{L}_n = \frac{1}{2} \sum_{i=1}^N \left[\log \sigma_\theta^2(\mathcal{G}_i) + \frac{\sum_{j=1}^K w_j \cdot (\mathcal{G}_{norm}^j - \mathcal{G}_i)^2 + C}{\sigma_\theta^2(\mathcal{G}_i)} \right], \quad (8)$$

where $w_j = \text{softmax}(-\log \sigma_\theta^2(\mathcal{G}_{norm}^j))$,

and the constant number C can be disregarded. The normalized weight w_j indicates the significance of $\mathcal{G}_{norm}^j$ in estimating $\mathcal{P}_{norm}$. Consequently, $\sum_{j=1}^K w_j \cdot (\mathcal{G}_{norm}^j - \mathcal{G}_i)^2$ quantifies the weighted distance between the current graph $\mathcal{G}_i$ and the estimated K normality-representative graphs. A lower value of $\sum_{j=1}^K w_j \cdot (\mathcal{G}_{norm}^j - \mathcal{G}_i)^2$ refers to greater similarity between the $\mathcal{G}_i$ and the estimated K normality-representative graphs, implying a higher probability that $\mathcal{G}_i$ is normal with a lower uncertainty $\sigma_\theta^2(\mathcal{G}_i)$. In contrast, the higher the distance $\sum_{j=1}^K w_j \cdot (\mathcal{G}_{norm}^j - \mathcal{G}_i)^2$, the higher the uncertainty $\sigma_\theta^2(\mathcal{G}_i)$ learned to attenuate potential anomalies.

However, a struggle remains for minimizing $\mathcal{L}_n$: How to calculate $(\mathcal{G}_{norm}^j - \mathcal{G}_i)^2$? There are several techniques in the literature for computing the distance between graph data $\mathcal{G}_{norm}^j$ and $\mathcal{G}_i$, such as graph kernels [52], graph edit distance [15], or distance mapping by deep vectorized space. Recent research [40, 60] has shown that Euclidean distances in vector space are well suited to show outlier

data. Hence, we employ GNNs to obtain vectorized representations of graph data and compute Euclidean distances, denoted as,

$$(\mathcal{G}_{norm}^j - \mathcal{G}_i)^2 = \|\mathbf{h}_{\mathcal{G}_{norm}^j} - \mathbf{h}_{\mathcal{G}_i}\|_2^2, \quad (9)$$

where $\mathbf{h}_{\mathcal{G}_{norm}^j}$ and $\mathbf{h}_{\mathcal{G}_i}$ are the GNN generated embeddings for graphs $\mathcal{G}_{norm}^j$ and $\mathcal{G}_i$, respectively. As mentioned in Section 3, the network for uncertainty learning consists of a GNN plus an MLP. Here, we adopt the GNN included in the uncertainty learner without an extra computational burden.

4.4 Potential Anomalies Exposure

The loss $\mathcal{L}_n$ in Eq. 8 is crafted to learn from potential normal graphs, maximizing the likelihood between normality and graphs with less uncertainty. For potential anomalies with high uncertainty, $\mathcal{L}_n$ attenuates their impacts but never exploits them. When being trained in a fully unsupervised setting, exploiting anomalies for model training seems a reasonable and promising direction. Therefore, we design a specific loss $\mathcal{L}_a$ to learn potential anomalies, derived from the opposite effects of the $\mathcal{L}_n$, as

$$\mathcal{L}_a = \frac{\sigma_\theta^2(\mathcal{G}_i)}{\sum_{j=1}^K w_j \cdot (\mathcal{G}_{norm}^j - \mathcal{G}_i)^2}, \quad (10)$$

where $w_j = \text{softmax}(-\log \sigma_\theta^2(\mathcal{G}_{norm}^j))$, and $\sum_{j=1}^K w_j \cdot (\mathcal{G}_{norm}^j - \mathcal{G}_i)^2$ evaluates the distance between the graph and normality. If graphs are considered potential anomalies due to their high uncertainty, $\mathcal{L}_a$ will push them away from normality. In addition, the optimization objective with $\mathcal{L}_n$ focuses solely on aligning all samples to a single distribution (normality), which can diminish the diversity of model representation, potentially risking feature collapse, i.e., the encoder maps all graphs into a single point or a degenerate subspace to minimize the loss. The $\mathcal{L}_a$ can alleviate this issue by acting as a penalty term to separate each sample from normality to varying degrees based on its uncertainty.

4.5 Final Objectives

The GNN encoder. As yet, we have $\mathcal{L}_n$ and $\mathcal{L}_a$ for training potential normal graphs and potential anomalies, respectively. However, a minor concern remains that the GNN encoder has not undergone specialized training. The GNN encoder plays a role in modeling uncertainty and calculating distances between graphs. To ensure model robustness, we employ the reconstruction loss of the Graph AutoEncoder (GAE) for training the GNN encoder,

$$\hat{\mathbf{A}} = \sigma(\mathbf{H}\mathbf{H}^\top), \quad \mathcal{L}_g = \frac{1}{N} \sum_{\mathcal{G} \in \mathbb{G}} \|\mathbf{A} - \hat{\mathbf{A}}\|_F^2, \quad (11)$$

where $\hat{\mathbf{A}}$ is the reconstructed adjacency matrix, and $\mathbf{H}$ is the node embedding matrix generated by the GNN.

The Overall Loss. Finally, we briefly review our GUL framework, including the E-step and M-step. GUL estimates the normality-representative graphs in the E-steps with fixed network parameter θ . In the M-step, GUL trains the network via the overall loss, as follows,

$$\mathcal{L} = \lambda_n \mathcal{L}_n + \lambda_a \mathcal{L}_a + \lambda_g \mathcal{L}_g, \quad (12)$$

where λ_n , λ_a , and λ_g are hyper-parameters to control weights of losses. The ablation study and the hyperparameter analysis of the proposed losses are presented in Sections 5.5 and 5.6, respectively.

During the inference stage, the uncertainty learned by GUL can be directly utilized to score the degree of abnormality of the input sample, as higher uncertainty indicates a greater probability of being an anomaly. To be notified, the time complexity analysis and pseudocode of our methods can be found in Appendix B, and the time efficiency is evaluated in Appendix C.4.

5 Experiments

5.1 Experimental Setup

Datasets. Following previous settings [26, 30–32, 35, 53], we evaluate GLAD detectors under a fully unsupervised setting with 16 widely used benchmark datasets, where samples from the minority or true anomalous class are treated as anomalies and the rest as normal. Our experiments are conducted in a fully unsupervised setting, utilizing the training data "as-is", i.e., retaining all normal and anomalous samples without any pre-filtering. More details and data statistics are provided in Appendix C.1.

Baselines. We adopt four baselines to enhance existing GLAD detectors under fully unsupervised scenarios, that are, 1) **Blind** trains GLAD detectors without performing any strategy under fully unsupervised settings. 2) **Refine** [54] filters out potential anomalous data based on a predefined ratio. 3) **LOE_H** [41] applies a separate anomaly-specific loss to filtered potential anomalies. 4) **LOE_S** [41] weights standard (normal) and specialized (anomaly) losses for each sample based on its predicted normality/anomaly probability, as a soft training constraint. 5) **IAD** [21] designs an importance weight update function based on the statistics of anomaly scores, to scale training loss. 6) **AutoOD** [6] filters out potential anomalies based on the mean and standard deviation of accumulated early training losses.

The ground-truth anomaly ratio is provided for these four baselines to achieve the best performance. These four baselines are applied with four backbone GLAD detectors, i.e., OCGIN [60], OCGTL [40], SIGNET [30], and MUSE [22].

Metric and Implementations. To follow previous work [2, 37], we report the mean and standard deviation (std) AUC metric across 5-fold cross-validation. In addition, GIN [49] is adopted as the GNN encoder. More details can be found in Appendix C.2.

5.2 Anomaly Detection Results under Fully Unsupervised Settings

The anomaly detection results in 16 datasets, shown in Table 1, yield three key observations: 1) SUL consistently outperforms other baselines, with a 5.91% average AUC gain, confirming the value of modeling uncertainty for fully unsupervised GLAD. 2) GUL significantly outperforms all baselines on almost all datasets, achieving substantial gains (e.g., 8.09% ~ 46% on MUTAG, $\geq 5.73\%$ on MMP) and a further average AUC gain of 6.80% over SUL, highlighting the advantage of direct uncertainty modeling from graph data. 3) Refine and LOE_H underperform the blind strategy in 54.7% and 45.2% of cases, respectively, due to over-reliance on potentially flawed detector predictions. In contrast, LOE_S applies a soft constraint and

underperforms in only 26.2% of cases, revealing that hard reliance on detector predictions can amplify bias.

5.3 Efficacy of Uncertainty Attenuation

As illustrated in Eq. 8, the learned uncertainty is adopted as a weighting factor, while higher uncertainty leads to a stronger loss term, attenuating the loss of potential anomalies. To validate this, Fig. 3 compares the uncertainty distributions of normal and anomalous graphs. Normal samples show low uncertainty (mainly 0.0–0.4), while anomalies shift toward higher uncertainty (0.6–1.0). This shift in the uncertainty distribution for anomalies indicates that the GUL effectively assigns higher uncertainty values to anomalous samples during training.

5.4 Anomaly Detection Results under Varying Contamination Ratios

We also evaluate the performance under increasing contamination ratios⁴ ranging from 10% to 100%, to investigate the robustness of baselines and our methods. A contamination ratio of 100% represents the fully unsupervised setting, where the training data is used "as-is" —that is, containing all inherent normal and anomalous samples. The backbone GLAD detector is set to OCGTL, which employs multiple GNNs with self-supervised learning to achieve great robustness. As shown in Fig. 4, two key observations emerge: 1) Our uncertainty-based methods (SUL and GUL) maintain strong performance across all contamination levels, with GUL benefiting further by leveraging $\mathcal{L}_a$ to learn from anomalies as contamination increases. 2) In contrast, Refine and LOE_H, which heavily rely on detector predictions, exhibit the steepest performance decline as contamination increases, eventually even underperforming the Blind strategy due to amplified bias.

5.5 Ablation Study

The ablation study highlights the necessity of losses in Eq. 12. As shown in Table 2, the three losses contribute distinctively to the model's performance: 1) $\mathcal{L}_n$ is essential to model normality, while its removal leads to major performance drops, such as 7.2% AUC decrease on MUTAG. 2) $\mathcal{L}_a$ is crucial for learning anomalous signals, especially in datasets with high anomaly ratios; removing it leads to AUC drops by 5.3% on NCI1. 3) $\mathcal{L}_g$ provides moderate but consistent improvements by constraining the GNN encoder, with the AUC decreasing between 1.0% and 2.0% when removed.

5.6 Hyper-parameter Sensitivity Analysis

Top-K graphs for presenting normality. In the E-step of GUL, we select the top-K graphs with the lowest uncertainty to approximate normality. Higher values of K enable a more comprehensive approximation of normality by integrating more graphs; however, this comes with the trade-off of higher computational costs and the potential introduction of noise. We tested $K \in \{3, 6, 12, 24, 48, 96, 198\}$ on four datasets (excluding $K = 198$ on MUTAG due to its small training size). As shown in Fig. 5, GUL is robust to the choice of K, with AUC fluctuations under 1.88%.

⁴In this paper, the contamination ratio (c%) means that we take all samples from the normal class mixed with c% of samples from the anomaly class for model training.

Table 1: Anomaly detection performance in terms of AUC (in percent, mean $\pm$ std) under fully unsupervised settings. Here we retain all normal and anomalous graphs in the training set without any removal. Gray highlighting denotes the best overall performance, while bold underlining identifies the best results within each GLAD detector.

Models		Datasets															
		MUTAG	DD	NCI1	PROTEINS	AIDS	BZR	COX2	DHFR	IMDB	REDDIT	HSE	MMP	p53	PPAR	BA-TYPE	Mutagen
OCGIN	Blind	50.95 $\pm$ 8.78	61.10 $\pm$ 1.95	48.16 $\pm$ 2.26	41.27 $\pm$ 4.34	99.05 $\pm$ 0.77	65.97 $\pm$ 3.34	65.95 $\pm$ 9.75	61.42 $\pm$ 4.43	53.91 $\pm$ 4.39	86.63 $\pm$ 2.14	64.55 $\pm$ 2.67	66.45 $\pm$ 1.15	65.23 $\pm$ 2.03	64.33 $\pm$ 4.22	79.26 $\pm$ 2.51	49.63 $\pm$ 1.61
	Refine	50.83 $\pm$ 6.65	61.23 $\pm$ 1.94	51.61 $\pm$ 1.73	42.81 $\pm$ 3.60	99.26 $\pm$ 0.54	64.25 $\pm$ 7.68	67.78 $\pm$ 9.15	61.44 $\pm$ 4.64	52.65 $\pm$ 3.83	88.00 $\pm$ 0.41	63.51 $\pm$ 1.01	66.80 $\pm$ 0.99	65.48 $\pm$ 1.85	63.69 $\pm$ 1.48	79.50 $\pm$ 3.21	48.20 $\pm$ 1.48
	LOE _H	50.95 $\pm$ 7.62	61.23 $\pm$ 1.94	43.84 $\pm$ 5.33	41.92 $\pm$ 3.38	99.23 $\pm$ 0.59	62.10 $\pm$ 8.58	67.91 $\pm$ 8.77	61.45 $\pm$ 4.66	53.69 $\pm$ 3.64	88.17 $\pm$ 1.43	65.22 $\pm$ 2.18	66.66 $\pm$ 1.35	65.45 $\pm$ 2.21	65.01 $\pm$ 5.14	81.97 $\pm$ 3.26	49.08 $\pm$ 1.55
	LOE _S	54.46 $\pm$ 8.68	63.78$\pm$2.86	43.12 $\pm$ 1.91	45.64 $\pm$ 3.92	99.16 $\pm$ 0.65	69.56 $\pm$ 3.23	66.99 $\pm$ 9.26	61.32 $\pm$ 4.81	54.11 $\pm$ 3.36	87.34 $\pm$ 1.77	65.25$\pm$2.30	66.53 $\pm$ 1.07	65.24 $\pm$ 2.36	64.56 $\pm$ 2.06	81.26 $\pm$ 3.16	47.50 $\pm$ 1.76
	IAD	52.95 $\pm$ 6.79	61.61 $\pm$ 3.36	54.51 $\pm$ 3.50	44.89 $\pm$ 3.76	99.16 $\pm$ 0.62	66.37 $\pm$ 8.96	66.94 $\pm$ 11.13	61.45 $\pm$ 4.99	54.16 $\pm$ 3.98	87.67 $\pm$ 1.44	65.13 $\pm$ 2.17	66.32 $\pm$ 3.00	65.43 $\pm$ 2.52	65.37 $\pm$ 2.98	81.79 $\pm$ 1.16	49.44 $\pm$ 1.59
	AutoOD	51.62 $\pm$ 5.91	61.92 $\pm$ 3.64	50.13 $\pm$ 2.77	43.78 $\pm$ 3.17	99.14 $\pm$ 0.65	65.95 $\pm$ 6.47	66.21 $\pm$ 8.39	61.51 $\pm$ 4.32	53.93 $\pm$ 4.16	86.82 $\pm$ 0.80	65.11 $\pm$ 2.73	66.58 $\pm$ 1.32	66.86 $\pm$ 3.60	65.15 $\pm$ 4.80	81.93 $\pm$ 3.50	50.35 $\pm$ 1.68
	SUL	84.12$\pm$3.38	62.38 $\pm$ 1.66	69.11$\pm$0.96	73.64$\pm$3.54	99.33$\pm$0.79	69.89$\pm$5.01	68.08$\pm$7.95	61.91$\pm$3.65	54.20$\pm$2.66	88.24$\pm$0.71	64.13 $\pm$ 0.19	66.82$\pm$1.43	67.27$\pm$1.27	66.23$\pm$1.06	82.38$\pm$2.04	57.48$\pm$1.73
OCGTL	Blind	70.34 $\pm$ 9.65	69.55 $\pm$ 2.48	61.26 $\pm$ 0.84	63.84 $\pm$ 3.25	96.50 $\pm$ 0.73	69.27 $\pm$ 7.64	63.33 $\pm$ 6.62	61.59 $\pm$ 4.68	56.33 $\pm$ 2.37	80.09 $\pm$ 1.29	62.96 $\pm$ 1.99	63.55 $\pm$ 1.86	67.93 $\pm$ 2.29	51.46 $\pm$ 5.55	88.65 $\pm$ 2.40	61.12 $\pm$ 3.39
	Refine	47.38 $\pm$ 7.27	63.32 $\pm$ 3.97	53.91 $\pm$ 1.05	43.66 $\pm$ 1.75	99.31$\pm$0.58	71.56 $\pm$ 6.51	66.50$\pm$7.34	61.53 $\pm$ 4.64	55.30 $\pm$ 2.93	80.55 $\pm$ 1.68	58.90 $\pm$ 2.06	64.31 $\pm$ 3.04	67.13 $\pm$ 2.87	50.42 $\pm$ 6.54	88.68 $\pm$ 2.63	58.48 $\pm$ 3.71
	LOE _H	48.00 $\pm$ 6.77	62.85 $\pm$ 3.41	52.71 $\pm$ 1.48	43.65 $\pm$ 2.04	99.25 $\pm$ 0.64	71.50 $\pm$ 7.35	66.13 $\pm$ 7.61	61.53 $\pm$ 4.65	55.11 $\pm$ 3.00	80.35 $\pm$ 1.92	62.37 $\pm$ 4.02	63.36 $\pm$ 3.41	66.63 $\pm$ 2.60	56.60 $\pm$ 5.51	89.04 $\pm$ 2.99	58.50 $\pm$ 3.74
	LOE _S	69.17 $\pm$ 10.59	71.66 $\pm$ 3.01	61.56 $\pm$ 1.62	60.27 $\pm$ 3.36	98.51 $\pm$ 0.78	71.48 $\pm$ 8.68	63.87 $\pm$ 6.37	61.54 $\pm$ 4.68	55.96 $\pm$ 2.99	81.42 $\pm$ 0.94	61.86 $\pm$ 1.61	64.03 $\pm$ 2.88	67.13 $\pm$ 3.78	50.53 $\pm$ 5.63	89.99 $\pm$ 2.25	59.34 $\pm$ 3.41
	IAD	71.02 $\pm$ 6.58	71.95 $\pm$ 3.31	62.35 $\pm$ 2.84	65.86 $\pm$ 1.98	99.01 $\pm$ 0.67	69.65 $\pm$ 7.24	63.87 $\pm$ 7.31	61.98$\pm$4.72	56.42 $\pm$ 2.71	81.32 $\pm$ 1.83	62.99 $\pm$ 1.88	64.01 $\pm$ 2.17	67.13 $\pm$ 3.55	55.60 $\pm$ 3.67	88.47 $\pm$ 2.56	61.15 $\pm$ 2.20
	AutoOD	72.69 $\pm$ 5.57	70.09 $\pm$ 3.55	61.98 $\pm$ 2.24	63.89 $\pm$ 2.80	98.51 $\pm$ 0.91	69.72 $\pm$ 5.75	65.26 $\pm$ 6.92	61.95 $\pm$ 4.84	55.79 $\pm$ 2.42	80.61 $\pm$ 1.47	62.98 $\pm$ 2.01	64.43 $\pm$ 5.11	68.01 $\pm$ 2.13	58.93 $\pm$ 1.58	87.54 $\pm$ 2.16	62.01 $\pm$ 1.58
	SUL	85.29$\pm$5.44	73.30$\pm$2.53	67.34$\pm$0.65	76.57$\pm$3.08	96.37 $\pm$ 0.79	72.38$\pm$4.67	64.31 $\pm$ 6.20	61.60 $\pm$ 2.25	56.60$\pm$3.47	81.60$\pm$2.48	64.66$\pm$1.71	65.68$\pm$1.47	68.22$\pm$1.99	62.34$\pm$3.95	92.93$\pm$3.47	63.51$\pm$1.66
SIGNET	Blind	61.17 $\pm$ 20.39	51.82 $\pm$ 4.42	48.37 $\pm$ 1.09	59.46 $\pm$ 6.86	66.82 $\pm$ 7.20	57.89 $\pm$ 8.04	55.92 $\pm$ 3.81	59.08 $\pm$ 3.08	60.20 $\pm$ 13.64	80.51 $\pm$ 1.85	56.83 $\pm$ 3.49	53.03 $\pm$ 4.57	52.10 $\pm$ 2.95	56.77 $\pm$ 4.65	72.96 $\pm$ 7.80	57.42 $\pm$ 15.11
	Refine	61.11 $\pm$ 4.37	47.56 $\pm$ 3.57	51.44 $\pm$ 7.50	50.36 $\pm$ 8.15	62.22 $\pm$ 10.94	55.06 $\pm$ 2.56	51.25 $\pm$ 8.10	57.18 $\pm$ 2.35	60.65 $\pm$ 3.59	75.22 $\pm$ 6.55	56.06 $\pm$ 2.65	54.01 $\pm$ 2.69	52.64 $\pm$ 2.81	56.34 $\pm$ 5.18	69.12 $\pm$ 1.40	54.33 $\pm$ 1.52
	LOE _H	67.32 $\pm$ 12.03	51.87 $\pm$ 7.09	51.86 $\pm$ 2.73	63.83 $\pm$ 4.86	71.21 $\pm$ 10.60	63.22 $\pm$ 5.79	51.27 $\pm$ 9.73	58.19 $\pm$ 5.01	59.83 $\pm$ 5.21	76.05 $\pm$ 3.44	55.96 $\pm$ 2.89	61.03 $\pm$ 4.99	57.34 $\pm$ 7.65	56.27 $\pm$ 5.31	69.31 $\pm$ 1.29	55.91 $\pm$ 9.00
	LOE _S	55.38 $\pm$ 20.16	52.48 $\pm$ 5.53	49.63 $\pm$ 2.87	64.61 $\pm$ 6.59	73.69 $\pm$ 7.30	64.34 $\pm$ 9.78	57.14 $\pm$ 13.08	61.69$\pm$3.09	61.38 $\pm$ 7.36	77.36 $\pm$ 1.86	56.49 $\pm$ 3.00	60.73 $\pm$ 6.50	54.72 $\pm$ 4.93	57.29 $\pm$ 5.13	73.35 $\pm$ 5.05	55.90 $\pm$ 4.76
	IAD	72.38 $\pm$ 13.34	54.70 $\pm$ 2.65	51.89 $\pm$ 2.63	59.88 $\pm$ 6.62	67.82 $\pm$ 7.20	58.89 $\pm$ 8.18	56.90 $\pm$ 3.79	59.09 $\pm$ 3.09	62.59 $\pm$ 8.87	80.83 $\pm$ 1.75	56.92 $\pm$ 3.24	58.89 $\pm$ 2.77	55.87 $\pm$ 2.33	57.99 $\pm$ 2.00	73.67 $\pm$ 9.95	58.88 $\pm$ 5.21
	AutoOD	72.06 $\pm$ 15.66	55.68$\pm$4.16	52.01 $\pm$ 2.72	60.87 $\pm$ 8.94	68.44 $\pm$ 8.01	59.80 $\pm$ 7.42	56.41 $\pm$ 5.13	59.34 $\pm$ 8.77	63.13 $\pm$ 9.28	79.37 $\pm$ 2.90	57.33 $\pm$ 4.81	60.21 $\pm$ 2.91	58.21 $\pm$ 4.52	58.13 $\pm$ 3.57	74.89 $\pm$ 8.23	57.51 $\pm$ 4.30
	SUL	75.51$\pm$13.50	54.28 $\pm$ 5.19	52.40$\pm$6.00	66.96$\pm$5.54	75.88$\pm$19.85	65.44$\pm$3.55	62.19$\pm$4.82	60.85 $\pm$ 10.20	65.10$\pm$4.97	81.36$\pm$1.23	58.36$\pm$1.02	62.16$\pm$2.45	63.04$\pm$1.31	61.92$\pm$1.81	76.58$\pm$2.75	62.63$\pm$11.78
MUSE	Blind	76.55 $\pm$ 9.02	68.70 $\pm$ 6.15	65.21 $\pm$ 1.84	73.66 $\pm$ 1.46	69.55 $\pm$ 4.79	68.06 $\pm$ 7.21	61.23 $\pm$ 8.03	62.37 $\pm$ 2.34	51.66 $\pm$ 2.97	60.72 $\pm$ 4.86	51.94 $\pm$ 2.60	56.04 $\pm$ 3.33	51.69 $\pm$ 1.25	54.33 $\pm$ 2.39	88.99 $\pm$ 2.36	62.02 $\pm$ 4.32
	Refine	64.18 $\pm$ 6.56	73.83 $\pm$ 2.66	65.03 $\pm$ 2.90	71.15 $\pm$ 4.23	66.88 $\pm$ 9.31	67.75 $\pm$ 8.12	61.20 $\pm$ 7.77	62.54 $\pm$ 2.07	51.22 $\pm$ 4.21	57.22 $\pm$ 7.81	51.16 $\pm$ 4.28	56.96 $\pm$ 2.31	52.90 $\pm$ 3.65	54.60 $\pm$ 2.06	88.11 $\pm$ 2.11	60.38 $\pm$ 3.11
	LOE _H	66.46 $\pm$ 2.30	72.10 $\pm$ 6.78	64.36 $\pm$ 1.40	74.41$\pm$2.78	58.43 $\pm$ 9.89	68.11 $\pm$ 6.18	59.86 $\pm$ 4.89	62.51 $\pm$ 9.58	52.10 $\pm$ 2.35	58.18 $\pm$ 15.27	52.99 $\pm$ 1.91	55.24 $\pm$ 4.72	52.08 $\pm$ 4.78	54.89 $\pm$ 4.37	83.52 $\pm$ 5.29	62.50 $\pm$ 3.54
	LOE _S	53.47 $\pm$ 16.28	75.53 $\pm$ 9.69	65.31 $\pm$ 1.67	73.90 $\pm$ 2.05	64.40 $\pm$ 9.13	68.91 $\pm$ 7.85	59.06 $\pm$ 3.95	62.73 $\pm$ 3.65	53.02 $\pm$ 2.96	57.89 $\pm$ 10.54	52.43 $\pm$ 3.85	56.66 $\pm$ 4.03	51.01 $\pm$ 3.50	53.70 $\pm$ 5.88	86.43 $\pm$ 4.50	63.34 $\pm$ 3.88
	IAD	82.35 $\pm$ 3.22	75.39 $\pm$ 4.73	64.39 $\pm$ 3.23	73.25 $\pm$ 7.91	74.22 $\pm$ 5.00	69.49 $\pm$ 8.93	56.88 $\pm$ 6.60	62.81 $\pm$ 4.23	51.83 $\pm$ 3.45	63.57 $\pm$ 8.99	52.15 $\pm$ 3.39	56.66 $\pm$ 3.21	52.88 $\pm$ 7.53	53.31 $\pm$ 2.21	89.77 $\pm$ 5.67	62.25 $\pm$ 3.14
	AutoOD	80.28 $\pm$ 3.87	74.24 $\pm$ 4.54	66.01 $\pm$ 2.72	73.58 $\pm$ 8.29	67.33 $\pm$ 3.37	69.43 $\pm$ 5.48	61.50 $\pm$ 6.48	62.74 $\pm$ 2.91	53.77 $\pm$ 4.12	65.51 $\pm$ 10.05	52.21 $\pm$ 6.32	56.77 $\pm$ 4.33	54.31$\pm$2.25	54.55 $\pm$ 4.55	89.03 $\pm$ 8.54	63.01 $\pm$ 1.55
	SUL	84.00$\pm$4.98	77.19$\pm$0.95	66.46$\pm$1.85	74.01 $\pm$ 0.88	77.03$\pm$9.40	69.96$\pm$7.19	62.90$\pm$6.82	63.30$\pm$3.31	54.89$\pm$2.18	70.47$\pm$5.25	54.43$\pm$1.76	57.87$\pm$3.80	52.72 $\pm$ 2.91	55.16$\pm$3.31	90.10$\pm$4.13	63.51$\pm$1.66
GUL		93.38 $\pm$ 1.02	78.13 $\pm$ 2.78	69.13 $\pm$ 2.20	79.56 $\pm$ 2.60	99.63 $\pm$ 0.33	75.28 $\pm$ 3.30	72.75 $\pm$ 6.77	63.87 $\pm$ 2.81	69.84 $\pm$ 5.90	83.66 $\pm$ 1.17	66.41 $\pm$ 5.26	72.55 $\pm$ 1.28	68.71 $\pm$ 1.29	67.94 $\pm$ 1.95	93.55 $\pm$ 1.45	70.64 $\pm$ 7.73

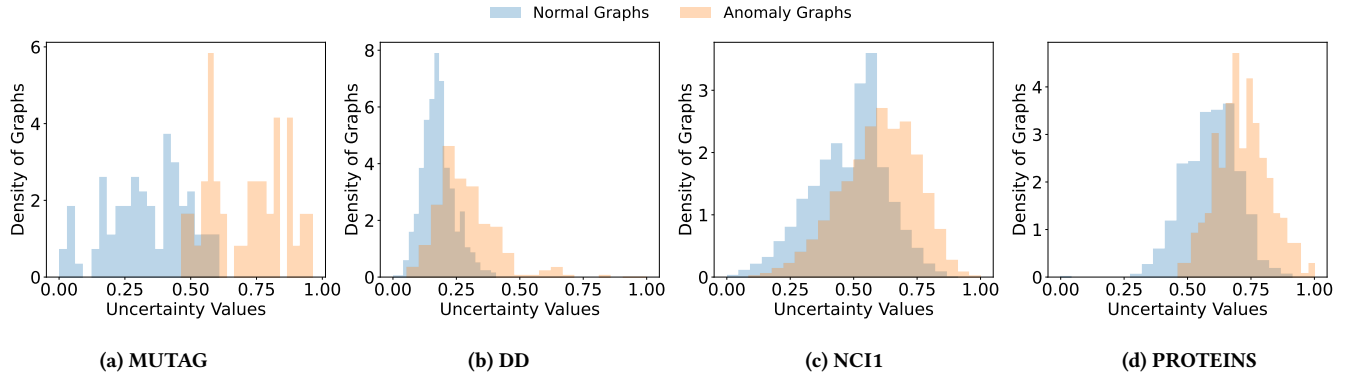


Figure 3: The density distributions of uncertainty for normal and anomaly samples during the GUL training. All uncertainty values have been normalized to the range $[0, 1]$ to ensure consistency.

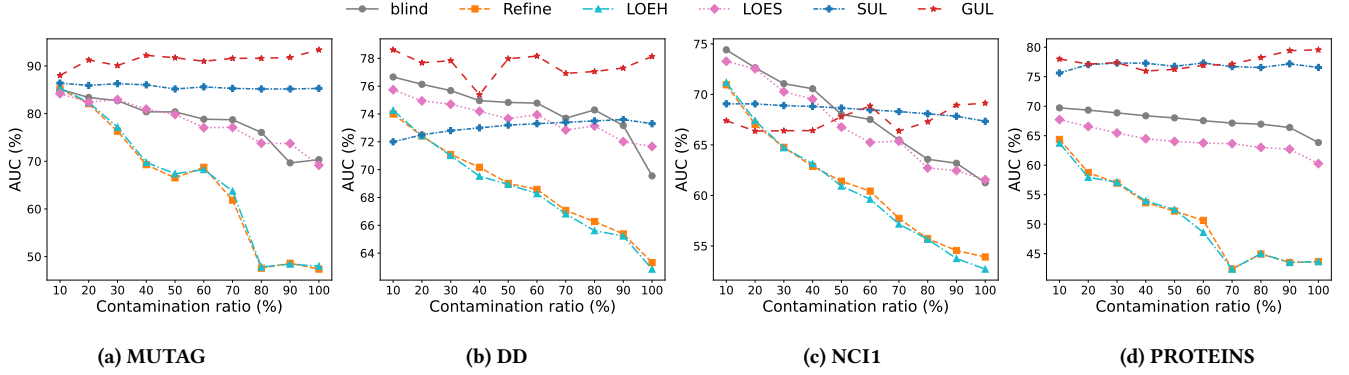


Figure 4: The AUC (%) performance of six strategies under increasing levels of contamination ratios (c%). The backbone GLAD detector, incorporating Blind, Refine, LOEH, LOES, and SUL, is OCGTL. The contamination ratio (c%) means that we take all normal graphs mixed with c% of graphs from the anomaly class for training. Notably, a contamination ratio of 100% represents the fully unsupervised setting where the training data is used "as-is", containing all inherent normal and anomalous graphs.

Table 2: Ablation Study for Proposed Loss.

$\mathcal{L}_n$	$\mathcal{L}_a$	$\mathcal{L}_g$	MUTAG	DD	NCI1	PROTEINS
✓	✓	✓	93.38±1.02	78.13±2.78	69.13±2.20	79.56±2.60
✓	✓	✓	86.18±3.51	69.64±2.03	63.07±5.65	70.64±3.96
✓	✓	✓	90.77±1.92	71.73±7.11	63.83±2.43	75.31±2.89
✓	✓	✓	92.37±3.28	76.94±2.63	67.20±2.52	78.25±2.38

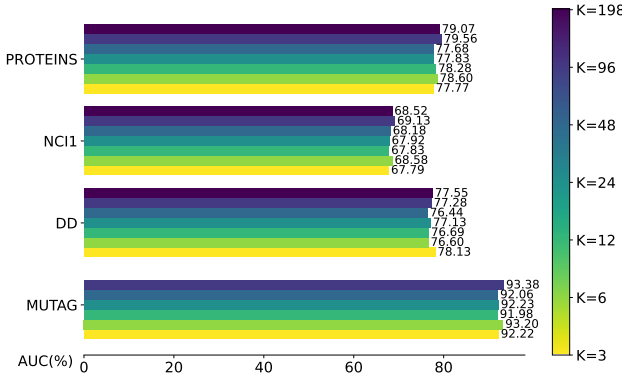


Figure 5: The AUC (%) performance of the GUL framework across varying values of K.

$\mathcal{L}_n$ and $\mathcal{L}_a$. The hyperparameters λ_n and λ_a balance learning from potential normal and anomalous samples. As shown in Fig. 6, GUL performs robustly across a wide range of values, especially when $\lambda_n, \lambda_a \in [0.4, 0.7]$. Notably, too low λ_n (e.g., ≤ 0.3) degrades performance, consistent with the ablation results in Section 5.5.

6 Conclusion

In this paper, we tackle the challenges of fully unsupervised graph-level anomaly detection (GLAD) by introducing uncertainty. Specifically, we propose two frameworks, Score Uncertainty Learning

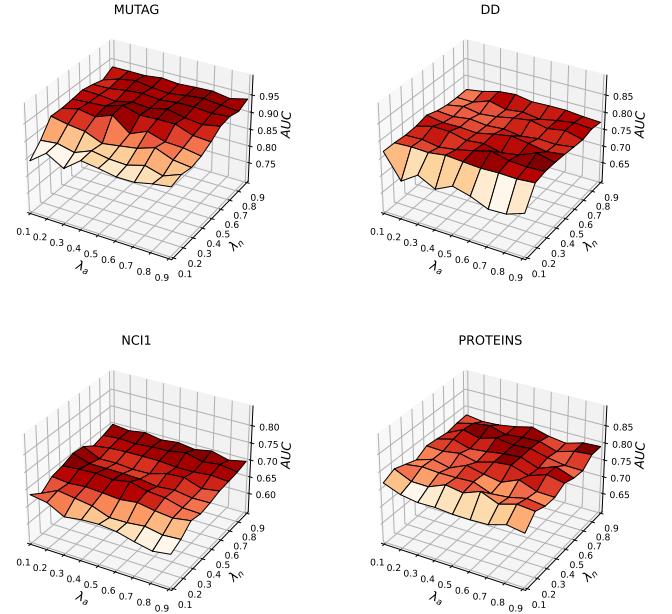


Figure 6: The AUC (%) performance of the GUL framework was evaluated across a range of values for λ_n and λ_a .

(SUL), which attenuates unreliable anomaly scores, and Graph-data Uncertainty Learning (GUL), which uses an EM algorithm to optimize anomaly detection and uncertainty modeling iteratively. Extensive experiments on sixteen benchmarks highlight the value of uncertainty in enhancing the reliability and effectiveness of GLAD detectors in fully unsupervised real-world scenarios.

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Algorithm 1: Graph-data Uncertainty Learning (GUL)

Input: The training set $\mathbb{G}_{\mathcal{T}}$ consisting of normal and anomalous graphs; Maximum iterations T ; Number of normality-representative graphs K ; Initial network parameter θ ; Loss weights $\lambda_n, \lambda_a, \lambda_g$.

Output: Trained parameters θ for modeling uncertainty and scoring abnormality.

```

1 Training: Initialize parameters  $\theta$ ;  $t = 1$ ;
2 while  $t \leq T$  do
3   E-step:
4   Compute uncertainty  $\sigma_{\theta}^2(\mathcal{G})$  for all graphs  $\mathcal{G} \in \mathbb{G}_{\mathcal{T}}$ ;
5   Select top- $K$  graphs with the lowest uncertainties to
   approximate normality  $\mathcal{G}_{norm} \leftarrow \text{Eq. 6}$ 
6   M-step:
7   for each graph  $\mathcal{G} \in \mathbb{G}_{\mathcal{T}}$  do
8     Compute uncertainty  $\sigma_{\theta}^2(\mathcal{G})$ ;
9     Compute losses  $\mathcal{L}_n$  and  $\mathcal{L}_a$  for potential normal and
       anomalous graphs, respectively, based on  $\mathcal{G}_{norm}$ 
       and  $\sigma_{\theta}^2(\mathcal{G}) \leftarrow \text{Eqs. 8 and 10}$ ;
10    Compute GAE loss  $\mathcal{L}_g \leftarrow \text{Eq. 11}$ ;
11    Back Propagation by minimizing
        $\mathcal{L} = \lambda_n \mathcal{L}_n + \lambda_a \mathcal{L}_a + \lambda_g \mathcal{L}_g \leftarrow \text{Eq. 12}$ ;
12    Update parameters  $\theta$ .
13    $t = t + 1$ ;
14 Inference:
15 for each input graph  $\mathcal{G}$  do
16   Compute uncertainty  $\sigma_{\theta}^2(\mathcal{G})$ ;
17   Higher  $\sigma_{\theta}^2(\mathcal{G})$  indicates high degree of abnormality.

```

A Supplementary of Related Work

Graph-level Anomaly Detection. Anomaly detection on graphs is a widely studied problem, ranging from anomalous nodes in a single graph [16, 17, 39, 44, 45], or anomalous graphs in a database/set of graphs. In a set of graphs, the majority typically exhibit stable and consistent patterns/properties, collectively called normality. Conversely, a small subset of graphs significantly deviates from this normality, identifying them as anomalies. Recently, GLAD has attracted growing interest from the deep graph learning community, prompting researchers to develop algorithms for detecting graph-level anomalies in both supervised and partially unsupervised manners. In the supervised setting, several studies [10, 28, 32, 43, 50, 57, 59] adapt graph classification models for the GLAD task, primarily by incorporating additional techniques, such as imbalanced classification. However, due to the scarcity of anomalous labels in practical applications, research has consequently shifted towards partially

unsupervised approaches that do not require labeled data, especially anomalies, for training. For instance, OCGIN [60] learns normality by mapping all normal data into a compact hypersphere in the encoded space, thereby enabling the encoder to map anomalies as outlier points. Building on this approach, OCGTL [40] incorporates transfer learning to enhance the robustness of the encoder. GlocalKD [31] learns normality by minimizing knowledge distillation error between a trainable network and a random network, where higher distillation error refers to higher probabilities of being anomalies. Similarly, SIGNET [30] aligns multi-view information via contrastive loss, with higher loss values indicating greater abnormality. In addition, MUSE [22] utilizes multifaceted summaries of reconstruction errors to learn robust normality for anomaly detection. These partially unsupervised GLAD methods train models exclusively using normal data to learn the normality via the loss functions, which are subsequently used to evaluate the degree of abnormality during inference. However, these methods are based on an overly stringent assumption, i.e., the exclusion of anomalies in the training set, which significantly hinders their applicability in real-world fully unsupervised scenarios.

Fully Unsupervised Anomaly Detection. Transitioning to a fully unsupervised setting introduces the challenge of training directly on data "as-is". To mitigate the impact of inherent anomalies, several general strategies have been proposed. Filtering-based methods, such as Refine [54] and AutoOD [6], attempt to purify the training set by removing potential outliers based on predefined ratios or early training loss statistics. And re-weighting strategies like LOE [41] and IAD [21] apply soft constraints or importance weights to down-weight potential anomalies, preventing them from biasing the training process. However, these strategies share critical limitations, such as their reliance on off-the-shelf detectors. These detectors often yield unreliable predictions on contaminated data, risking exacerbated bias rather than mitigation. Moreover, the requirement for prior knowledge, e.g., contamination ratios, further hinders their practical deployment. Alternatively, several EM-based approaches [3, 47] treat sample-wise anomaly labels as latent variables for iterative refinement. In contrast, our method defines the latent variable as the estimation of normality. This formulation is inherently more aligned with the essence of anomaly detection; it enables the model to robustly fit the underlying distribution, rather than undertaking the significantly harder and more error-prone task of sample-wise label inference.

Uncertainty Learning. Extensive research has long been conducted on the nature of uncertainties to support reliability evaluation and risk-based decision-making processes [9, 12, 38]. In general, the concept of uncertainty can be divided into two categories [1, 13]: 1) *model uncertainty*, which captures noise arising from network parameters, and 2) *data uncertainty*, which stems from noise inherently present in the data observations, such as label noise and low-quality data. In the context of fully unsupervised GLAD, the primary occurrence is *data uncertainty*. This is because existing GLAD detectors typically assume that all training graphs are normal, leading to the mislabeling of anomalies. These mislabeled anomalies carry inherent noise into the model's observations, leading to high uncertainty and poor performance. Current techniques, mostly focus on investigating *model uncertainty*, like Bayesian network [13, 14, 18–20] and ensemble learning [24, 29]. Several work

Table 3: The Statistical Information of Datasets. We explicitly list the class index (0 or 1) treated as anomalies and their corresponding semantic meanings for reproducibility.

Data	# Graphs(Norm/Anom)	Anom. Class ID	Anomaly Semantics	# Graphs(Train/Test)	Avg. #Nodes	Avg. #Edges
MUTAG	125/63	1	Mutagenic Effect	150/38	17.93	19.79
DD	691/487	0	Non-Enzyme Proteins	942/236	284.32	715.66
NCI1	2057/2053	1	Active (Lung Cancer)	3288/822	29.87	32.3
PROTEINS	663/450	0	Non-Enzyme Proteins	890/223	39.06	72.82
AIDS	1600/400	1	Active (Anti-HIV)	1600/400	15.69	16.2
BZR	319/86	0	Active (BZR Ligand)	324/81	35.75	38.36
COX2	365/102	0	Active (COX2 Inhibitor)	373/94	41.22	43.45
DHFR	461/295	1	Active (DHFR Inhibitor)	604/152	42.43	44.54
IMDB	500/500	1	Movie Genre (Sci-Fi)	800/200	19.77	96.53
REDDIT	1000/1000	1	Thread Type (Q&A)	1600/400	429.63	497.75
HSE	7722/428	0	Toxic (Human Health)	6520/1630	16.72	17.04
MMP	6178/1142	0	Toxic (MMP Disruptor)	5856/1464	17.49	17.83
p53	8097/537	0	Toxic (p53 Activator)	6907/1727	17.79	18.19
PPAR	7962/222	0	Toxic (PPAR Antagonist)	6547/1637	17.23	17.55
BA-TYPE	900/100	0	Diamond/5-Cycle Motifs	800/200	40.00	42.09
Mutagen	1936/1015	1	Mutagenic (-NO ₂ /-NH ₂)	2360/591	28.25	58.28

estimates *data uncertainty* in image [7, 56] and text [48] by utilizing label information. However, their success cannot extend to GLAD, as modeling uncertainty for irregular graph-structured data in label-scarce anomaly detection remains unresolved.

B Time Complexity Analysis and Algorithm Pseudocode

For each graph, the time complexity of SUL is $O(|\mathcal{E}|)$, where $|\mathcal{E}|$ is the number of edges within the graph. The primary computational cost of SUL arises from the GNN encoder, with a complexity of $O(|\mathcal{E}|)$. When plugged into GNN-based GLAD detectors, SUL can share their GNN layers to achieve better lightweight.

The time complexity of GUL is $O(|\mathcal{E}| + |\mathcal{V}|^2)$ for each input graph, where $|\mathcal{V}|$ denotes the number of nodes in the graph. Specifically, the GNN encoder contributes a computational cost of $O(|\mathcal{E}|)$, while the GAE loss $\mathcal{L}_g$ adds the cost of $O(|\mathcal{V}|^2)$. In addition, GUL incurs $O(K * N)$ computational burden for calculating $\mathcal{L}_n$ and $\mathcal{L}_a$, and $O(N \log N)$ complexity for sorting graphs based on uncertainty values. For a better understanding, the pseudocode of GUL is described in Algorithm 1.

C Supplementary of Experiments

C.1 Details of Datasets

To comprehensively evaluate the performance of GLAD detectors under fully unsupervised settings, we conduct experiments on sixteen benchmark datasets. These datasets have been widely used in GLAD tasks [26, 30–32, 35, 40, 60], involving biological proteins, chemical molecules, social networks, etc. In addition, to create a more challenging scenario, the BA-TYPE [53] and Mutagen [8, 30] datasets feature a mix of two or more anomaly types within the training set. Specifically, anomalies in BA-TYPE are synthetic graphs involving either a diamond or a 5-cycle motif, while those in Mutagen encompass mutagenic molecules characterized by the presence of nitro (-NO₂) or amino (-NH₂) groups.

Our experiments are conducted in a fully unsupervised setting, meaning we depart from previous studies by utilizing the training data "as-is" — retaining all normal and anomalous samples without any pre-filtering. The statistics of the used datasets are shown in Table 3.

C.2 Details of Implementations

Metrics. In this paper, the performance of the GLAD detector is evaluated by the most widely used metric in anomaly detection and GLAD, i.e., AUC [2, 37]. We use a training ratio of 20% and report the performance across 5-fold cross-validation.

Hyperparameters. We perform grid searches and experimental analysis in Section 5.6 for the hyperparameter setting. All experiments are implemented on Python 3.10.0, CUDA 11.6.1, a Rocky Linux 8.6 server with a 12-core CPU, 1 NVIDIA V100 GPU, and 300GB RAM. We set the learning rate to 0.001, the dropout rate to 0.3, and the number of training epochs to 1000. The remaining key hyperparameters of GUL are selected via grid search. Specifically, the hyperparameter search is performed over the following spaces:

- Number of layers within GNN encoder: {2,3,4,5};
- Dimension of GNN encoder: {16,32,64,128,256};
- λ_n , λ_a , and λ_g : {0.1,0.2,0.3,0.4,0.5,0.6,0.7,0.8,0.9};
- Value of K : {3,6,12,24,48,96,198}.

For fair comparisons, we also grid searched the key parameters for other baselines, including the layer numbers, learning ratio, hidden dimensions, etc. Some specific parameters within models are considered, such as the number of transformations in OCGTL.

C.3 Comprehensive Evaluation with PR-AUC

To address the concern regarding class imbalance in anomaly detection benchmarks, we further evaluate all methods using the Area Under the Precision-Recall Curve (PR-AUC). The results, reported in Table 4, align with the AUC observations and yield two key findings: 1) Across four backbone architectures, SUL achieves the best

Table 4: Anomaly detection performance in terms of PR-AUC (in percent, mean $\pm$ std) under fully unsupervised settings. Here we retain all normal and anomalous graphs in the training set without any removal. Gray highlighting denotes the best overall performance, while bold underlining identifies the best results within each GLAD detector.

Models		Datasets															
		MUTAG	DD	NCI1	PROTEINS	AIDS	BZR	COX2	DHFR	IMDB	REDDIT	HSE	MMP	p53	PPAR	BA-TYPE	Mutagen
OCGIN	Blind	41.00 $\pm$ 6.32	58.79 $\pm$ 1.87	50.82 $\pm$ 1.80	36.13 $\pm$ 2.54	98.51 $\pm$ 0.84	36.59 $\pm$ 7.00	40.92 $\pm$ 10.50	48.41 $\pm$ 4.17	55.36 $\pm$ 3.30	85.21 $\pm$ 2.33	7.91 $\pm$ 0.50	23.20 $\pm$ 0.83	9.90 $\pm$ 1.13	4.55 $\pm$ 0.81	51.00 $\pm$ 5.38	31.23 $\pm$ 1.07
	Refine	37.20 $\pm$ 5.62	59.78 $\pm$ 1.26	49.92 $\pm$ 1.68	40.29 $\pm$ 4.03	98.83 $\pm$ 0.62	29.05 $\pm$ 5.38	40.90 $\pm$ 9.33	48.08 $\pm$ 6.85	51.34 $\pm$ 2.87	86.50 $\pm$ 1.14	8.29 $\pm$ 0.71	23.19 $\pm$ 0.57	9.85 $\pm$ 0.80	4.93 $\pm$ 0.68	65.97 $\pm$ 4.89	30.37 $\pm$ 1.05
	LOE _H	35.95 $\pm$ 5.31	60.59 $\pm$ 1.21	43.26 $\pm$ 2.14	35.63 $\pm$ 2.39	98.70 $\pm$ 0.76	27.39 $\pm$ 5.51	39.54 $\pm$ 8.12	48.08 $\pm$ 3.88	53.02 $\pm$ 2.38	87.38 $\pm$ 1.24	7.78 $\pm$ 0.87	23.23 $\pm$ 0.75	9.64 $\pm$ 0.76	4.56 $\pm$ 0.73	66.54 $\pm$ 4.96	30.38 $\pm$ 1.05
	LOE _S	40.55 $\pm$ 9.49	60.70 $\pm$ 1.63	47.24 $\pm$ 1.86	36.72 $\pm$ 3.12	98.62 $\pm$ 0.79	36.24 $\pm$ 6.36	41.50 $\pm$ 10.49	48.51 $\pm$ 3.92	54.53 $\pm$ 3.24	86.78 $\pm$ 2.13	7.72 $\pm$ 0.70	23.16 $\pm$ 0.47	9.73 $\pm$ 1.02	5.09$\pm$1.37	60.44 $\pm$ 6.90	31.01 $\pm$ 1.01
	IAD	41.88 $\pm$ 6.82	59.21 $\pm$ 1.32	54.13 $\pm$ 3.09	36.32 $\pm$ 2.35	98.72 $\pm$ 0.73	38.32 $\pm$ 4.75	40.32 $\pm$ 10.30	48.26 $\pm$ 4.06	53.59 $\pm$ 3.13	87.43 $\pm$ 1.49	8.47 $\pm$ 0.84	23.17 $\pm$ 0.83	9.90 $\pm$ 1.17	4.62 $\pm$ 1.26	59.87 $\pm$ 5.31	31.23 $\pm$ 1.07
	AutoOD	38.57 $\pm$ 3.39	58.99 $\pm$ 1.49	52.11 $\pm$ 1.27	35.86 $\pm$ 1.86	98.52 $\pm$ 0.76	39.67 $\pm$ 7.24	40.27 $\pm$ 10.73	48.44 $\pm$ 4.21	57.33 $\pm$ 3.00	86.55 $\pm$ 1.13	8.30 $\pm$ 0.97	23.10 $\pm$ 0.81	9.87 $\pm$ 1.09	4.80 $\pm$ 0.87	60.40 $\pm$ 7.59	32.21 $\pm$ 1.05
	SUL	76.37$\pm$4.93	63.92$\pm$3.64	64.34$\pm$1.35	67.99$\pm$5.83	99.21$\pm$0.95	42.45$\pm$9.90	41.55$\pm$6.30	48.94$\pm$1.95	58.31$\pm$3.40	87.54$\pm$0.88	8.55$\pm$2.43	23.30$\pm$0.78	10.21$\pm$1.87	4.92 $\pm$ 1.72	68.72$\pm$7.08	33.05$\pm$1.18
OCGTL	Blind	57.33 $\pm$ 6.70	59.68 $\pm$ 4.20	57.97 $\pm$ 0.78	50.70 $\pm$ 3.17	93.87 $\pm$ 1.08	47.04 $\pm$ 14.62	40.99 $\pm$ 9.09	47.95 $\pm$ 3.49	54.75 $\pm$ 2.79	80.63 $\pm$ 2.17	8.68 $\pm$ 1.34	20.64 $\pm$ 1.88	9.68 $\pm$ 1.11	5.02 $\pm$ 3.52	62.23 $\pm$ 6.09	35.21 $\pm$ 1.51
	Refine	33.24 $\pm$ 3.70	49.14 $\pm$ 3.06	47.92 $\pm$ 0.53	35.87 $\pm$ 1.25	98.74 $\pm$ 0.82	46.24 $\pm$ 9.73	42.06$\pm$7.97	47.93 $\pm$ 3.43	55.07 $\pm$ 1.91	80.96 $\pm$ 1.29	8.83 $\pm$ 1.21	21.25 $\pm$ 0.84	9.19 $\pm$ 1.29	4.62 $\pm$ 1.48	69.91 $\pm$ 7.90	37.57 $\pm$ 1.75
	LOE _H	33.13 $\pm$ 3.68	48.45 $\pm$ 2.74	47.90 $\pm$ 0.83	35.85 $\pm$ 1.49	98.75$\pm$0.80	47.02 $\pm$ 10.86	41.55 $\pm$ 7.87	47.90 $\pm$ 3.42	55.28 $\pm$ 1.97	81.25 $\pm$ 2.40	7.86 $\pm$ 0.85	20.79 $\pm$ 0.79	9.18 $\pm$ 0.74	5.02 $\pm$ 2.64	66.58 $\pm$ 2.36	37.59 $\pm$ 1.75
	LOE _S	52.99 $\pm$ 0.97	57.67 $\pm$ 3.53	58.20 $\pm$ 3.47	46.76 $\pm$ 2.95	97.25 $\pm$ 0.92	46.24 $\pm$ 9.93	41.68 $\pm$ 9.00	47.91 $\pm$ 3.47	54.16 $\pm$ 2.06	79.45 $\pm$ 2.06	8.85 $\pm$ 1.89	21.45 $\pm$ 0.67	9.41 $\pm$ 1.17	4.51 $\pm$ 1.94	62.65 $\pm$ 6.05	35.59 $\pm$ 1.59
	IAD	59.53 $\pm$ 4.08	49.67 $\pm$ 2.86	57.98 $\pm$ 0.84	45.48 $\pm$ 1.19	98.51 $\pm$ 0.91	46.64 $\pm$ 8.05	36.46 $\pm$ 8.02	48.12 $\pm$ 3.40	55.13 $\pm$ 1.91	81.09 $\pm$ 2.21	7.22 $\pm$ 0.77	20.32 $\pm$ 0.87	9.67 $\pm$ 1.66	3.67 $\pm$ 1.03	62.14 $\pm$ 9.06	35.23 $\pm$ 1.52
	AutoOD	60.13 $\pm$ 6.80	52.91 $\pm$ 3.21	58.43 $\pm$ 2.59	47.41 $\pm$ 2.75	96.94 $\pm$ 0.76	41.68 $\pm$ 8.92	39.77 $\pm$ 8.41	47.95 $\pm$ 3.49	55.08 $\pm$ 2.81	78.95 $\pm$ 2.16	8.65 $\pm$ 1.32	21.39 $\pm$ 0.91	9.45 $\pm$ 0.84	4.46 $\pm$ 2.71	60.71 $\pm$ 4.24	40.21 $\pm$ 5.20
	SUL	77.25$\pm$5.16	64.61$\pm$3.25	62.11$\pm$1.40	69.56$\pm$5.02	95.38 $\pm$ 4.39	48.10$\pm$12.61	39.57 $\pm$ 7.92	48.97$\pm$2.59	56.99$\pm$2.83	81.52$\pm$2.21	8.86$\pm$0.89	22.88$\pm$1.04	10.23$\pm$1.02	5.09$\pm$2.06	71.06$\pm$5.87	43.52$\pm$3.30
SIGNET	Blind	59.17 $\pm$ 14.41	51.30 $\pm$ 4.78	48.04 $\pm$ 0.70	50.57 $\pm$ 6.59	28.78 $\pm$ 10.32	30.42 $\pm$ 8.54	31.29 $\pm$ 11.48	48.57 $\pm$ 7.42	60.85 $\pm$ 10.96	78.23 $\pm$ 4.40	5.39 $\pm$ 1.27	18.21 $\pm$ 0.89	8.61 $\pm$ 1.32	3.01 $\pm$ 0.99	53.45 $\pm$ 26.52	41.87 $\pm$ 5.90
	Refine	56.84 $\pm$ 7.64	52.14 $\pm$ 3.51	51.97 $\pm$ 2.78	47.58 $\pm$ 7.70	27.73 $\pm$ 13.59	27.07 $\pm$ 10.51	28.30 $\pm$ 9.44	48.69 $\pm$ 6.58	60.56 $\pm$ 6.69	76.36 $\pm$ 2.06	5.92 $\pm$ 1.36	19.20 $\pm$ 1.13	8.99 $\pm$ 0.88	3.89 $\pm$ 1.71	50.99 $\pm$ 8.48	36.49 $\pm$ 11.99
	LOE _H	59.26 $\pm$ 19.81	52.11 $\pm$ 3.53	51.87 $\pm$ 2.31	55.05 $\pm$ 8.40	36.04 $\pm$ 8.80	37.11 $\pm$ 14.29	27.96 $\pm$ 10.14	49.26 $\pm$ 2.44	59.58 $\pm$ 3.74	77.28 $\pm$ 3.11	5.11 $\pm$ 1.08	19.97 $\pm$ 1.22	9.52 $\pm$ 1.21	3.90 $\pm$ 1.21	54.65 $\pm$ 16.05	41.16 $\pm$ 10.51
	LOE _S	50.19 $\pm$ 16.05	53.20 $\pm$ 3.82	49.62 $\pm$ 1.99	54.87 $\pm$ 3.18	33.82 $\pm$ 10.40	37.47 $\pm$ 11.70	32.75 $\pm$ 8.77	49.57 $\pm$ 6.79	58.56 $\pm$ 8.42	76.54 $\pm$ 3.33	6.02 $\pm$ 1.11	19.88 $\pm$ 1.35	9.03 $\pm$ 1.23	4.33 $\pm$ 1.30	55.52$\pm$10.30	41.66 $\pm$ 13.47
	IAD	61.26 $\pm$ 14.65	52.32 $\pm$ 4.51	51.04 $\pm$ 0.70	50.40 $\pm$ 4.12	29.77 $\pm$ 10.33	30.05 $\pm$ 8.09	31.28 $\pm$ 11.47	48.54 $\pm$ 7.73	60.64 $\pm$ 10.79	78.51 $\pm$ 1.65	5.48 $\pm$ 1.25	18.87 $\pm$ 2.00	9.11 $\pm$ 0.97	4.72 $\pm$ 2.68	53.40 $\pm$ 28.92	43.52 $\pm$ 5.62
	AutoOD	60.10 $\pm$ 14.38	52.33 $\pm$ 2.56	52.24 $\pm$ 0.75	52.65 $\pm$ 7.94	29.48 $\pm$ 10.95	29.64 $\pm$ 8.42	32.46 $\pm$ 7.85	49.76$\pm$4.59	60.90 $\pm$ 11.05	78.43 $\pm$ 7.57	6.21 $\pm$ 1.53	19.96 $\pm$ 1.79	10.01 $\pm$ 0.52	4.39 $\pm$ 1.20	54.19 $\pm$ 31.05	40.16 $\pm$ 5.34
	SUL	67.19$\pm$8.48	53.51$\pm$4.99	55.11$\pm$0.87	62.23$\pm$5.21	39.25$\pm$18.04	40.03$\pm$6.33	39.52$\pm$5.87	49.67 $\pm$ 8.31	60.92$\pm$3.95	79.25$\pm$3.21	7.01$\pm$1.83	20.86$\pm$0.83	10.80$\pm$0.98	5.00$\pm$1.28	55.15 $\pm$ 3.48	45.86$\pm$10.21
MUSE	Blind	73.53 $\pm$ 6.63	63.01 $\pm$ 4.71	62.50 $\pm$ 0.51	52.46 $\pm$ 9.34	40.22 $\pm$ 13.08	53.16 $\pm$ 10.98	39.13 $\pm$ 12.96	47.01 $\pm$ 4.23	55.79 $\pm$ 4.03	61.23 $\pm$ 3.35	4.68 $\pm$ 1.27	18.96 $\pm$ 1.01	8.11 $\pm$ 0.52	3.76 $\pm$ 1.28	58.59 $\pm$ 8.93	42.02 $\pm$ 5.53
	Refine	58.86 $\pm$ 8.16	64.66 $\pm$ 2.39	60.31 $\pm$ 1.55	60.89 $\pm$ 7.29	41.69 $\pm$ 12.69	52.78 $\pm$ 12.42	35.71 $\pm$ 10.01	48.17 $\pm$ 6.22	54.72 $\pm$ 3.18	57.11 $\pm$ 2.52	4.98 $\pm$ 1.36	19.11 $\pm$ 1.33	8.37 $\pm$ 1.98	3.89 $\pm$ 1.40	63.90 $\pm$ 17.08	38.89 $\pm$ 10.03
	LOE _H	62.24 $\pm$ 3.68	63.25 $\pm$ 2.83	60.11 $\pm$ 3.51	58.80 $\pm$ 0.09	26.03 $\pm$ 15.97	52.98 $\pm$ 12.24	35.71 $\pm$ 4.48	47.93 $\pm$ 4.05	55.00 $\pm$ 2.94	56.01 $\pm$ 10.35	5.02 $\pm$ 1.33	18.61 $\pm$ 0.87	8.55 $\pm$ 0.81	3.92 $\pm$ 1.52	65.52 $\pm$ 13.65	39.11 $\pm$ 3.39
	LOE _S	54.64 $\pm$ 3.17	62.90 $\pm$ 5.95	59.98 $\pm$ 3.31	62.72 $\pm$ 11.74	29.42 $\pm$ 15.77	53.12 $\pm$ 11.18	34.49 $\pm$ 8.83	47.89 $\pm$ 5.33	56.34 $\pm$ 3.81	52.48 $\pm$ 12.21	4.88 $\pm$ 1.10	19.34 $\pm$ 0.98	8.03 $\pm$ 0.93	4.01 $\pm$ 1.33	65.47 $\pm$ 12.22	43.01 $\pm$ 4.41
	IAD	72.45 $\pm$ 7.21	63.30 $\pm$ 4.31	62.30 $\pm$ 3.55	60.91 $\pm$ 9.14	43.17$\pm$12.95	52.97 $\pm$ 10.24	37.20 $\pm$ 9.20	48.58 $\pm$ 5.82	52.26 $\pm$ 3.81	65.54 $\pm$ 5.62	4.97 $\pm$ 0.99	19.68 $\pm$ 0.79	8.99 $\pm$ 1.59	4.26 $\pm$ 0.80	62.62 $\pm$ 4.56	42.21 $\pm$ 5.68
	AutoOD	80.89 $\pm$ 6.84	63.37 $\pm$ 3.44	61.43 $\pm$ 2.58	61.56 $\pm$ 9.22	39.32 $\pm$ 21.97	53.10 $\pm$ 12.38	38.68 $\pm$ 4.17	48.51 $\pm$ 4.16	55.21 $\pm$ 4.83	64.81 $\pm$ 8.92	5.11 $\pm$ 1.36	19.95 $\pm$ 0.93	9.37$\pm$1.30	4.03 $\pm$ 1.51	63.83 $\pm$ 16.33	43.00 $\pm$ 4.03
	SUL	83.59$\pm$4.68	64.68$\pm$3.79	63.33$\pm$2.37	64.38$\pm$9.38	37.28 $\pm$ 14.31	54.11$\pm$5.10	39.67$\pm$1.39	49.12$\pm$7.31	57.40$\pm$5.20	69.98$\pm$4.57	5.76$\pm$1.80	20.03$\pm$0.90	9.02 $\pm$ 1.35	5.01$\pm$1.29	70.07$\pm$10.00	43.52$\pm$6.98
GUL		84.61$\pm$2.55	72.89$\pm$3.44	64.59$\pm$2.05	70.17$\pm$2.90	99.24$\pm$0.75	43.60$\pm$8.70	43.32$\pm$10.14	50.51$\pm$3.74	60.98$\pm$5.95	79.18$\pm$1.16	8.91$\pm$1.98	23.97$\pm$7.08	10.83$\pm$1.81	5.22$\pm$0.50	72.05$\pm$4.44	48.25$\pm$8.58

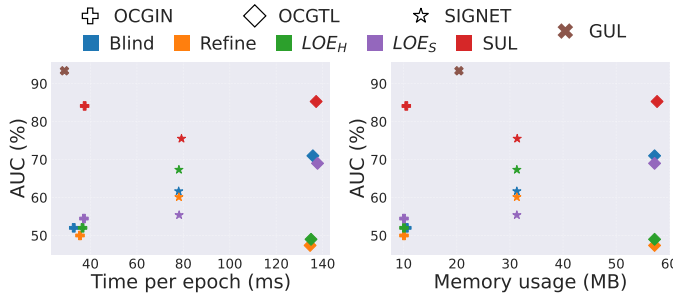


Figure 7: Efficiency analysis on MUTAG dataset.

performance among all baselines in most cases (bold underlined), confirming that modeling score-based uncertainty effectively mitigates the impact of anomalies even in imbalanced scenarios. 2) By directly modeling uncertainty from graph data, GUL outperforms all baselines on the vast majority of datasets (14 out of 16), securing

the best overall performance (gray highlighting), such as an 11.02% improvement over the best baseline on DD.

C.4 Efficiency Analysis

We report the running time per epoch and memory usage on graphs cards within the MUTAG dataset to verify efficiency. As shown in Fig. 7, the SUL demonstrates comparable time and memory costs to other baseline strategies while significantly improving backbone detectors. In addition, GUL achieves the fastest runtime and low memory usage but delivers state-of-the-art performance. These results highlight the efficiency and superior performance of our proposed methods. While generally efficient, GUL’s scalability on large graphs is constrained by the $O(V^2)$ complexity of the GAE loss in Eq. 11. To enhance scalability, future work can approximate the loss via edge sampling, substantially reducing computational overhead without sacrificing learning efficacy.